Assignment 2 - Git & GitHub Fundamentals
Group Report

- ***Group Size:*** 3 students
- ***Chosen Theme:*** Unit Converter
- ***Full Names:*** Aymen Nhaila, Douaa Moutawakkil, Malak Baddi
- ***Student IDs:*** Aymen Nhaila - 159632; Douaa Moutawakkil - 159621; Malak Baddi - 158934
- ***Course Codes & Sections:*** CSC 3324 - Aymen: Section 01; Douaa: Section 02; Malak: Section 03
- ***GitHub Usernames:*** @lurantys, @douaamoutawakkildd-svg, @m4lak-b
- ***GitHub Repository URL:*** https://github.com/lurantys/unit-converter
- ***GitHub Pages URL:*** https://lurantys.github.io/unit-converter/

1. Project & Role Distribution

The group followed the three-member path and chose the Unit Converter theme. Aymen Nhaila developed temperature conversion, Douaa Moutawakkil developed length conversion, and Malak Baddi developed weight conversion. The public repository contains the Python conversion modules, pytest tests, README, requirements file, GitHub Actions workflow, and a static project page.

2. Branches, Commits & Contributions

Aymen Nhaila: On *feature/temperature*, Aymen implemented Celsius, Fahrenheit, and Kelvin conversion and validation, then added five tests in two commits. Aymen also authored *docs/update-team-roles* (PR #3) and authored the README status update on *docs/update-project-status* (PR #8, approved by Malak and merged).

Douaa Moutawakkil: On *feature/length-conversion*, Douaa implemented conversions among meters, kilometers, centimeters, and miles and added five tests in two commits. Douaa also resolved the README conflict on *docs/status-length* (PR #4), added the Pages site on *feature/github-pages* (PR #6), and updated its visible version on *feature/pages-update* (PR #7).

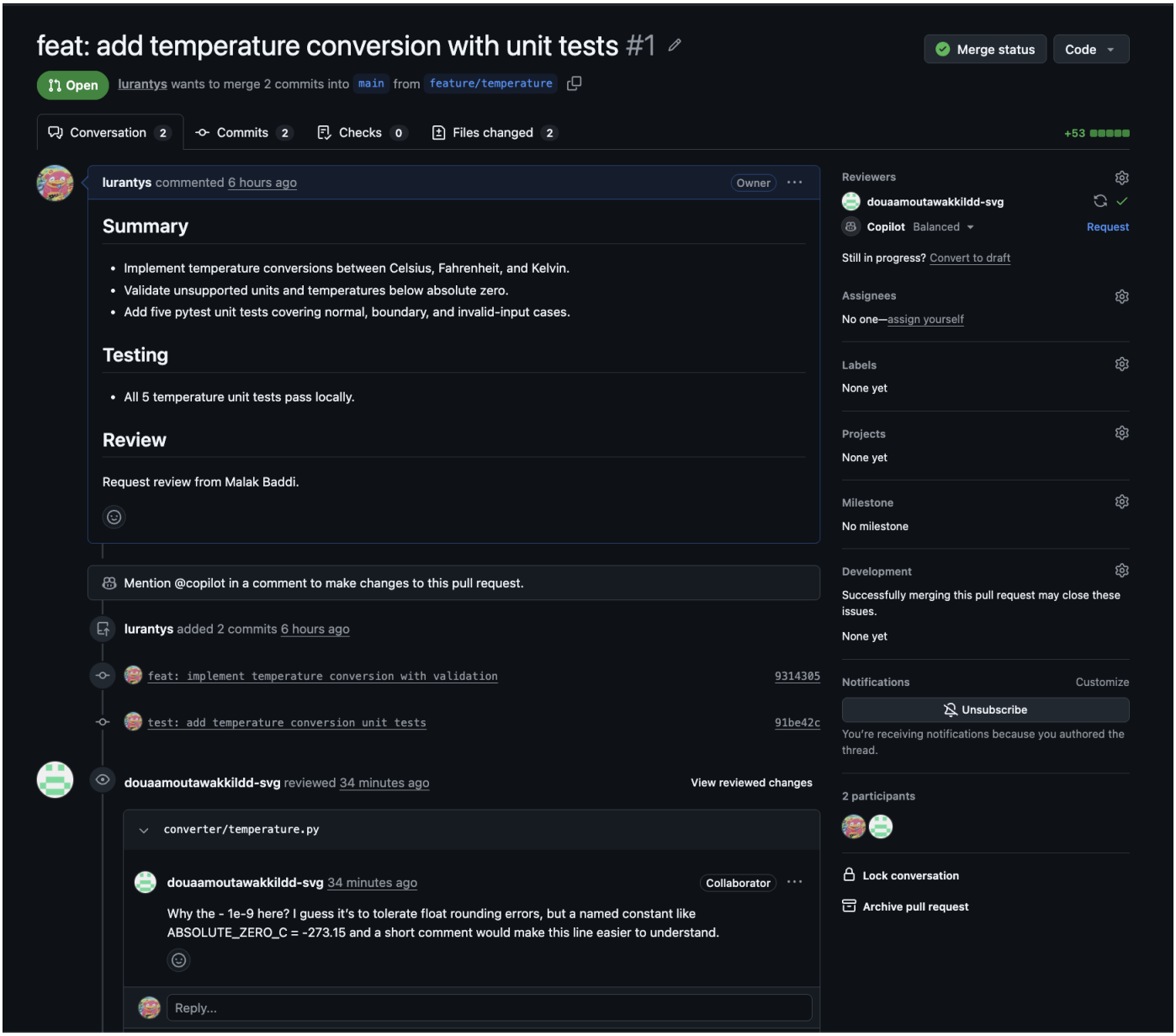
Malak Baddi: On *feature/weight-conversion*, Malak implemented kg, g, and lb conversions with conversion factors and validation for negative values, unsupported units, nonnumeric values, and booleans. She added nine tests and the CI workflow in six meaningful commits, including a failing-test demonstration and its correction.

3. Pull Requests & Code Review

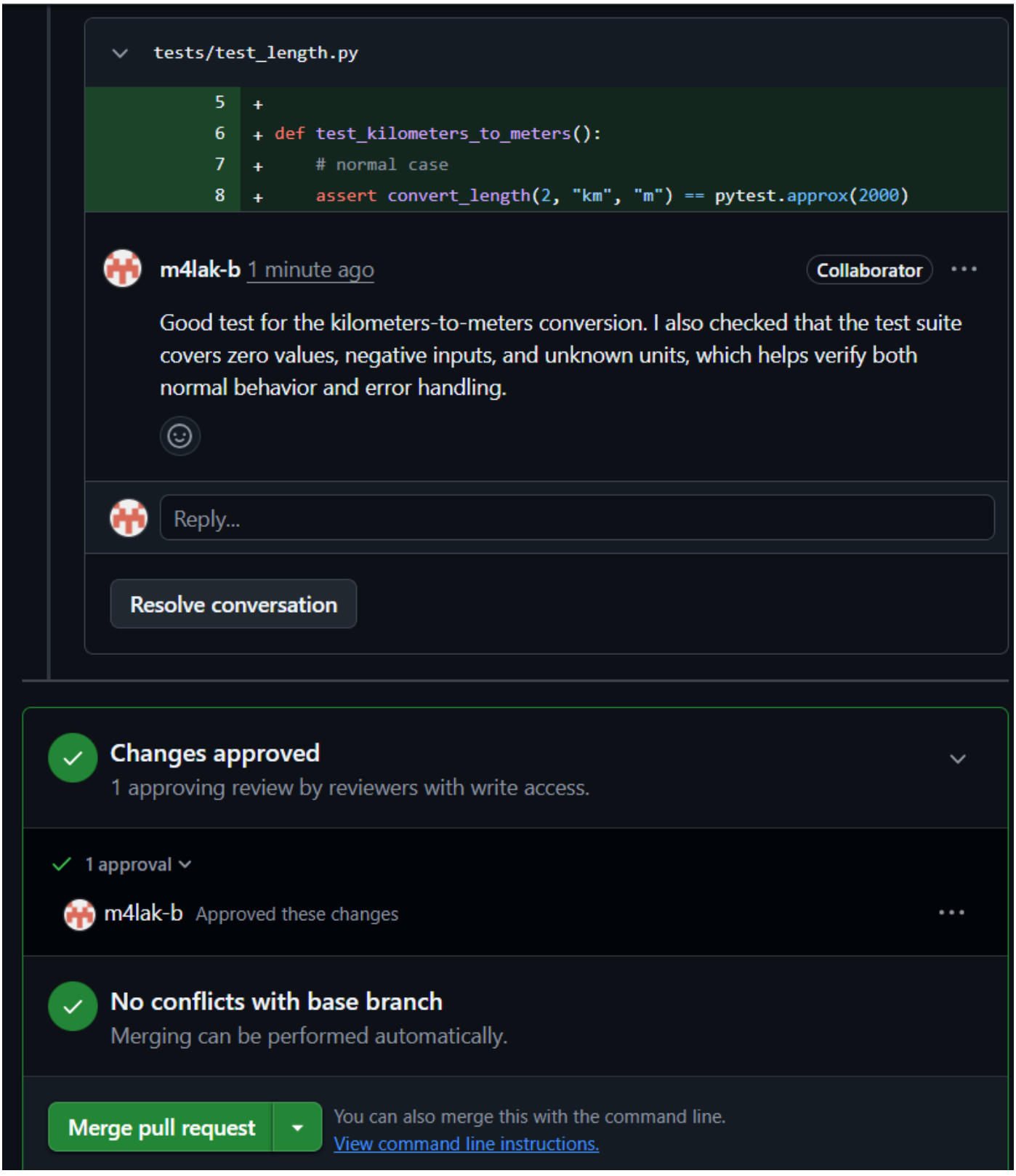
PR #1 (Aymen, temperature) received line-level feedback and approval from Douaa. PR #2 (Douaa, length) received review feedback and approval from Malak. PR #3 (Aymen, README roles) was approved by Douaa. PR #4 (Douaa, conflict resolution) was approved by Malak. PR #5 (Malak, weight and CI) received Request changes and later approval from Aymen. PR #6 (Douaa, Pages setup) and PR #7 (Douaa, website update) were approved by Aymen. PR #8 was approved by Malak and merged.

On PR #5, Aymen requested tests for nonnumeric and boolean inputs. Malak added them on the same branch in commit c234e69. Aymen reviewed the update and approved it. Checks passed before merge.

Douaa's review on PR #1 included a meaningful line comment and an approval:



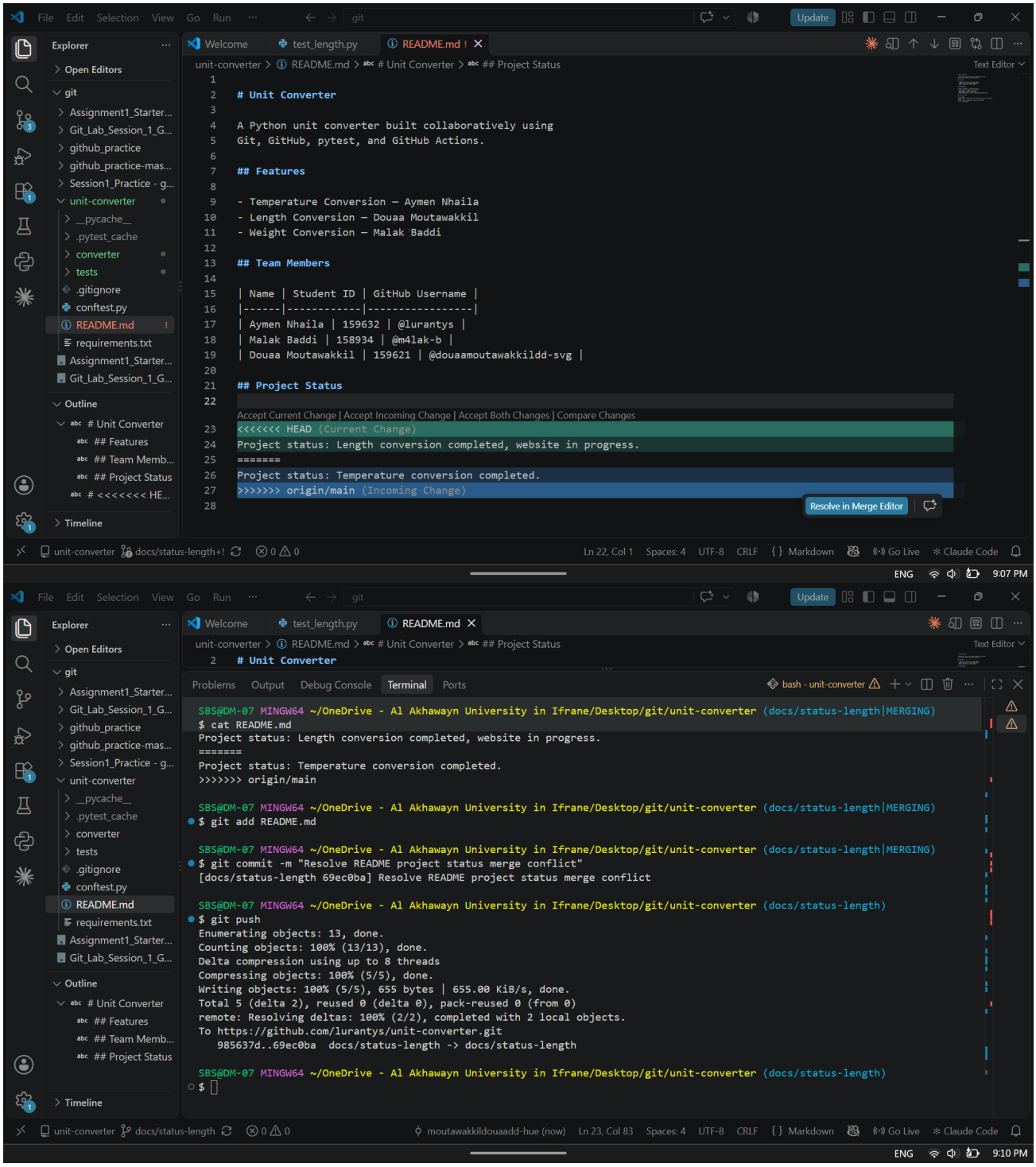
Malak reviewed the length conversion in PR #2, commented on the kilometer-to-meter test, and approved the changes:


tests/test_length.py
5 +
6 + def test_kilometers_to_meters():
7 + # normal case
8 + assert convert_length(2, "km", "m") == pytest.approx(2000)
m4lak-b 1 minute ago
Collaborator
Good test for the kilometers-to-meters conversion. I also checked that the test suite covers zero values, negative inputs, and unknown units, which helps verify both normal behavior and error handling.
Reply...
Resolve conversation
Changes approved
1 approving review by reviewers with write access.
1 approval
m4lak-b Approved these changes
No conflicts with base branch
Merging can be performed automatically.
Merge pull request
You can also merge this with the command line.
View command line instructions.

4. Merge Conflict

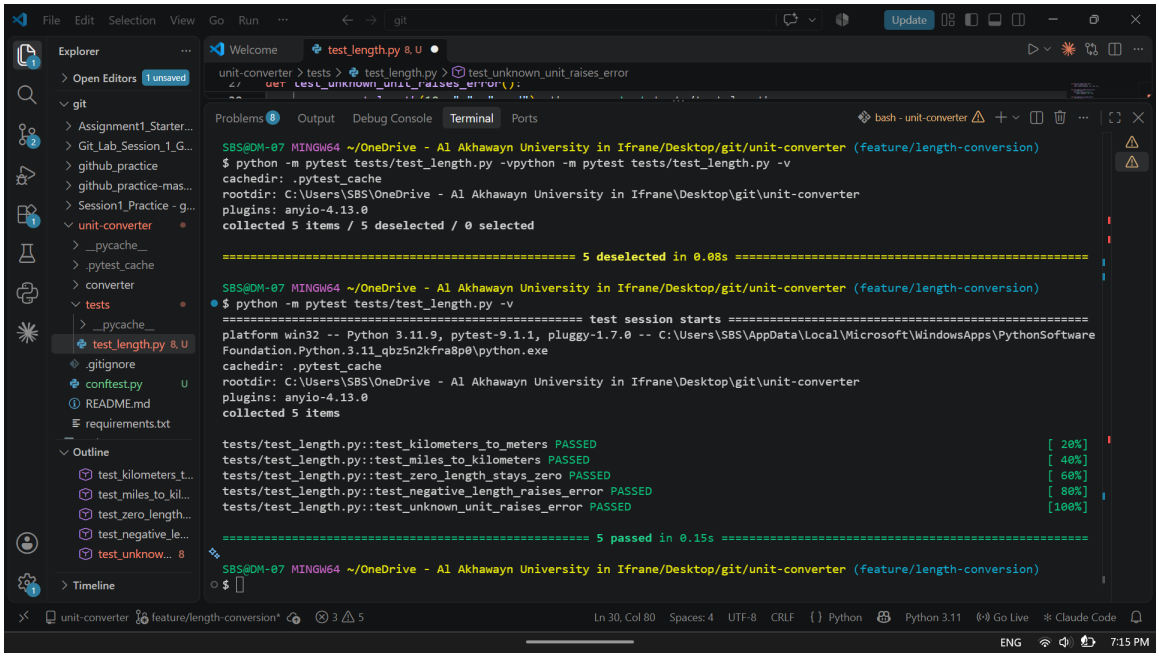
The group changed the same README project-status line on two branches. “Current Change” was Douaa’s length-progress wording on *docs/status-length*; “Incoming Change” was Aymen’s temperature-progress wording from *main*. Douaa combined both milestones as “Temperature and length conversions completed, website in progress.”, committed and pushed the resolution, and completed PR #4. Malak reviewed and approved the conflict-resolution PR.

The two branch versions and the resolved result are shown here:



5. Unit Tests

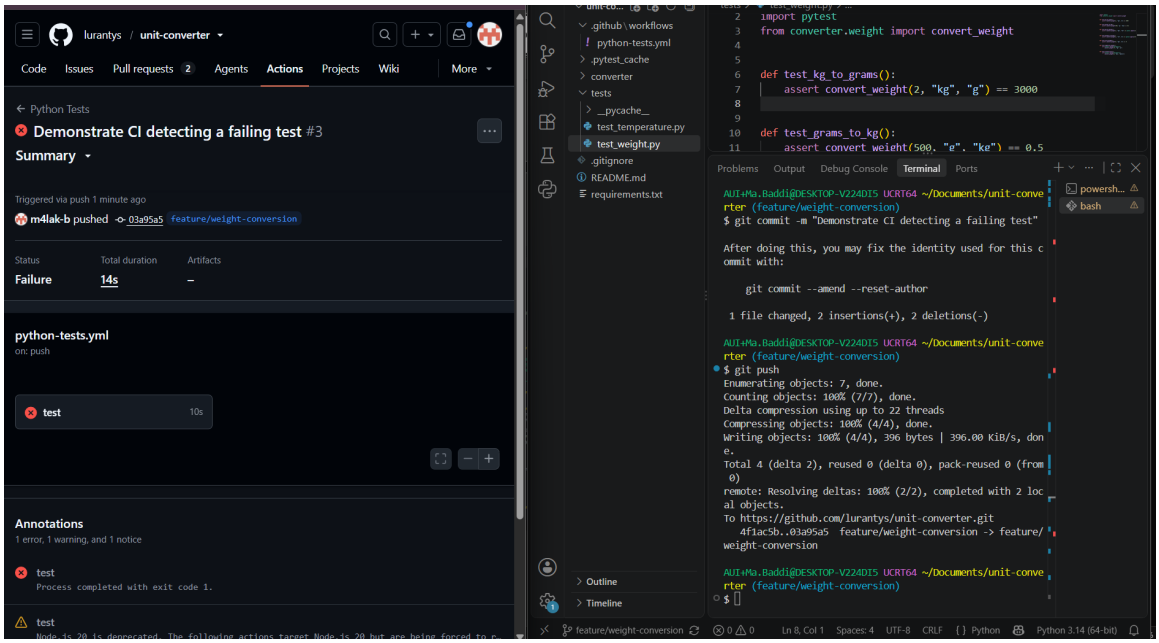
The final suite has 19 tests: five temperature, five length, and nine weight tests. The nine weight tests cover kg-to-g, g-to-kg, kg-to-lb, lb-to-g, zero, negative values, invalid units, nonnumeric input, and boolean input. Across the suite, tests also cover the absolute-zero boundary and normal, edge, and invalid cases. The local run shown below passed all five temperature tests.



6. Continuous Integration

The Python Tests workflow runs on pushes and when a pull request is opened or updated. It checks out the repository, uses an Ubuntu runner and Python 3.11, installs requirements.txt, and runs *python -m pytest -v*. An early weight test failed because it expected 3,000 g for 2 kg; the correct result is 2,000 g. Malak corrected the expectation in commit 574300b, after which CI passed. Later PR checks, including the review-requested tests, also passed.

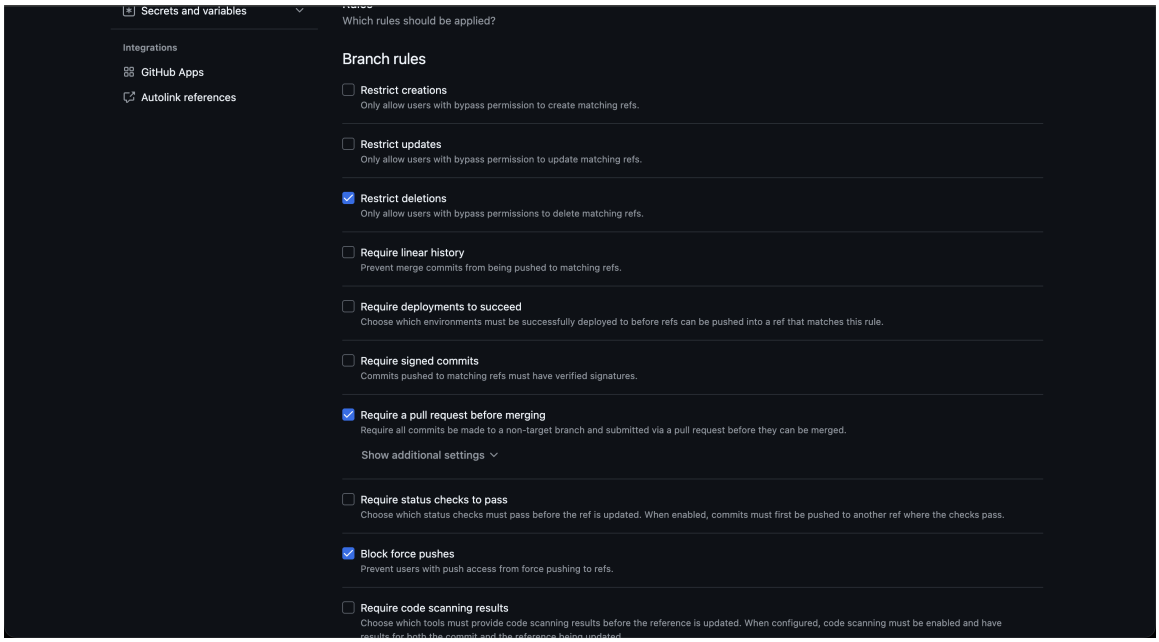
The failed run during development is shown here:



7. Ruleset & Protected main

The active “Protect main” ruleset targets the default branch and has an empty bypass list. It requires pull requests and one approving review, restricts branch deletion, and blocks force pushes. “Require status checks to pass” is unchecked and the ruleset has no status-check rule. CI runs and passes, but it is not enforced before merging. The assignment requirement to require both approval and successful CI is therefore incomplete.

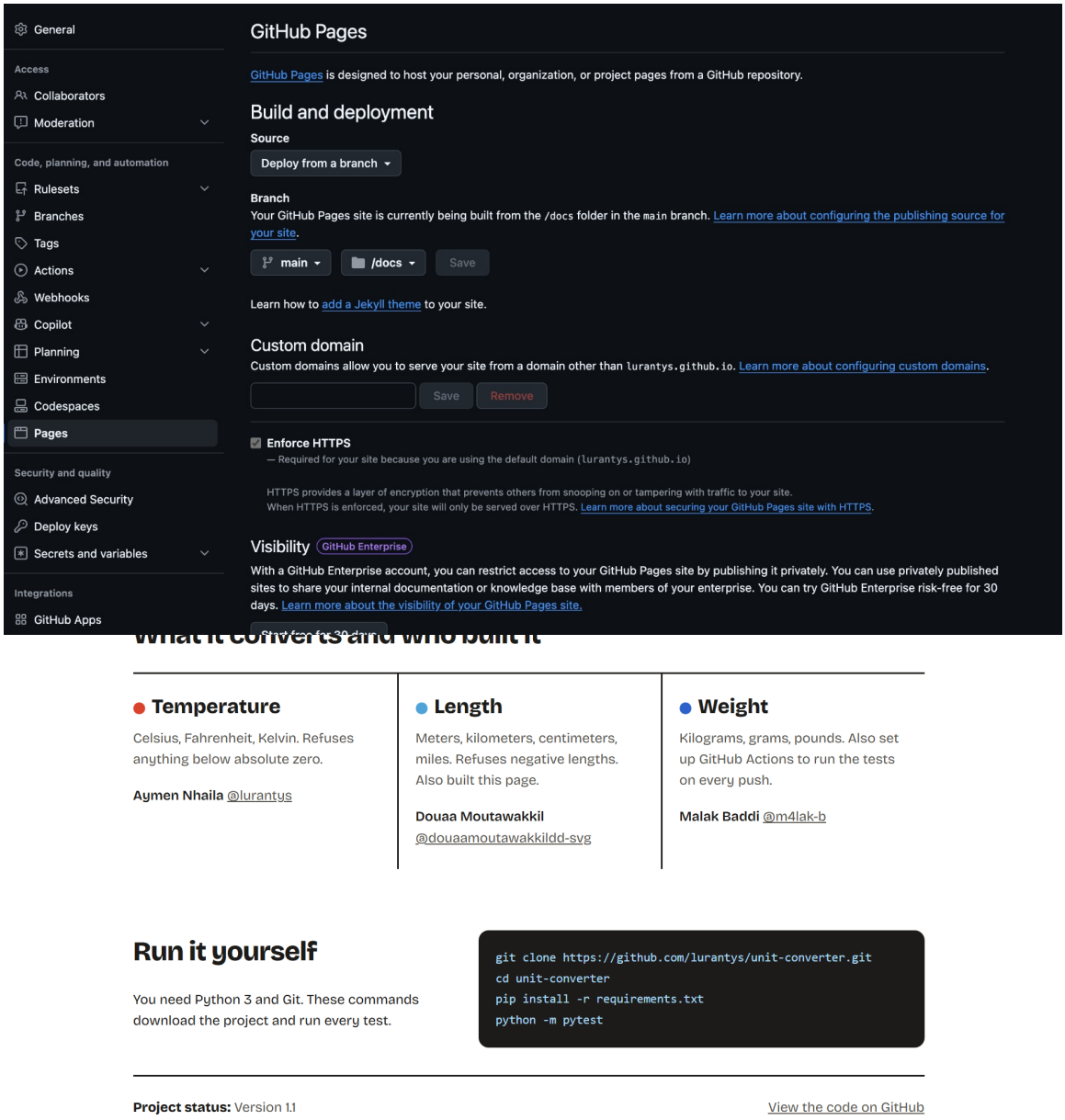
The visible ruleset settings are shown here:



8. Continuous Deployment

GitHub Pages publishes from the *main* branch’s */docs* folder. Douaa added the site in PR #6 and updated the visible status to Version 1.1 in PR #7. The deployment succeeded and the live page shows the team’s three conversion areas and run instructions.

Pages is configured to publish from main /docs, and the live page shows the update:



PR #8 was approved by Malak and merged. The main README now includes the live GitHub Pages URL.

9. Contribution Summary

Aymen Nhaila - Temperature, *feature/temperature*, PR #1; README roles, PR #3; status update, PR #8 (merged). Reviewed PR #5, requested changes, then approved after the same-branch correction. Added five temperature tests.

Douaa Moutawakkil - Length, *feature/length-conversion*, PR #2; conflict resolution, PR #4; Pages, PRs #6 and #7. Reviewed PR #1 with a line comment and approval; approved PR #3. Added five length tests.

Malak Baddi - Weight conversion and CI, *feature/weight-conversion*, PR #5. Reviewed PR #2 with feedback and approval, and approved PR #4. Added nine weight tests for conversions, zero, and invalid input including nonnumeric and boolean values.

10. Fair Use of AI

All three members used AI for advice, instructions, and debugging issues. The group used suggestions as guidance and checked the resulting work through tests, CI results, pull-request reviews, and the published site. Each member remains responsible for understanding the code and collaboration steps attributed to them.

11. Sources & Reflection

Sources: CSC 3324 Assignment 2 handout; Malak’s weight-conversion and CI contribution summary; project repository, https://github.com/lurantys/unit-converter; live site, https://lurantys.github.io/unit-converter/; GitHub documentation for repository rulesets and GitHub Pages; pytest documentation.

The README conflict was the hardest collaboration problem because two branches changed the same shared line. The group kept both updates by combining them into one status sentence, then reviewing the resolution before merge. The deliberately failing weight test showed why expected results should be checked independently. If a pull request or workflow behaves unexpectedly, first inspect the changed files, exact failed check or assertion, branch and base state, and active ruleset.

Before submission, enable the Python Tests status check in the main ruleset. The main README now includes the live Pages URL. The group must also add this PDF to the repository and submit the repository URL in the shared spreadsheet.